Math 401 — Applications of Linear Algebra

Exercise Solutions

Least Squares · Curve Fitting · Team Ranking · Graph Theory · Markov Chains · Google PageRank

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Chapter 1

Least Squares — Solutions

Exercise 3.1. Find the least-squares solution and error for $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 5 & 1 & 1 \\ 2 & 2 & 0 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$.

Solution. Normal equations $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$:

$$A^T A = \begin{pmatrix} 31 & 11 & 6 \\ 11 & 9 & 10 \\ 6 & 10 & 14 \end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix} 24 \\ 13 \\ 10 \end{pmatrix}.$$

```
A = [1 2 3; -1 0 2; 5 1 1; 2 2 0]; b = [1;2;3;4];
x_hat = inv(A'*A)*A'*b;  ls_error = norm(A*x_hat-b);
```

Output: $\hat{\mathbf{x}} \approx (0.3619,\ 0.9281,\ 0.0951)^T$, error ≈ 3.0035. □

Exercise 3.2. Find the vector in $\mathrm{Col}(A)$ closest to $\mathbf{b}$ for $A = \begin{pmatrix} 1 & 2 \\ 0 & -3 \\ 2 & 6 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

Solution. $A^T A = \begin{pmatrix} 5 & 14 \\ 14 & 49 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, $\det = 49$.

$$\hat{\mathbf{x}} = \frac{1}{49}\begin{pmatrix} 49 & -14 \\ -14 & 5 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 9/7 \\ -19/49 \end{pmatrix}.$$

Closest vector: $A\hat{\mathbf{x}} = \begin{pmatrix} 25/49 \\ 57/49 \\ 12/49 \end{pmatrix} \approx \begin{pmatrix} 0.510 \\ 1.163 \\ 0.245 \end{pmatrix}$. □

Exercise 3.3. Find the vector in $\mathrm{span}\left\{\begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \\ 0 \\ 3 \end{pmatrix}\right\}$ closest to $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$.

Solution. $A = \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ -1 & 0 \\ 0 & 3 \end{pmatrix}$. Normal equations: $A^T A = \begin{pmatrix} 6 & 6 \\ 6 & 27 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$. $\det = 162 - 36 = 126$. $\hat{\mathbf{x}} = \begin{pmatrix} -1/3 \\ 2/9 \end{pmatrix}$. Closest vector: $A\hat{\mathbf{x}} = \begin{pmatrix} 1/3 \\ 0 \\ 1/3 \\ 2/3 \end{pmatrix}$. □

Exercise 3.4. Can $A^T A\mathbf{x} = A^T\mathbf{b}$ have no solutions?

Solution. **No.** The projection $\mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ always exists, so the normal equation always has at least one solution. □

Exercise 3.5. Can more variables than equations give a unique least-squares solution?

Solution. **No.** If A is $m \times n$ with $n > m$, the columns of A are linearly dependent (at most m can be independent). Dependent columns $\Rightarrow$ infinitely many least-squares solutions. □

Exercise 3.6. Find the vector in $\mathrm{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}$ closest to $\begin{pmatrix}5\\5\end{pmatrix}$.

Solution. $A = \begin{pmatrix}1\\2\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}5\\5\end{pmatrix}$. Normal equation: $5\hat{t} = 15 \Rightarrow \hat{t} = 3$. Closest vector: $\begin{pmatrix}3\\6\end{pmatrix}$. □

Exercise 3.7. Find the point on $y = 3x$ closest to $(2, 3)$.

Solution. Matrix form: $\begin{pmatrix}1\\3\end{pmatrix} t = \begin{pmatrix}2\\3\end{pmatrix}$. Normal equation: $10\hat{t} = 11 \Rightarrow \hat{t} = 1.1$. **Closest point:** $(1.1,\ 3.3)$. □

Exercise 3.8. Show the least-squares formula gives the exact solution when A is invertible.

Solution. $(A^T A)^{-1} A^T\mathbf{b} = A^{-1}(A^T)^{-1}A^T\mathbf{b} = A^{-1}\mathbf{b}$. This is the exact solution. □ □

Exercise 3.9. Show $x + 2y = 4$, $x + 2y = 3$ has no unique least-squares solution.

Solution. $A = \begin{pmatrix}1 & 2\\1 & 2\end{pmatrix}$ has dependent columns. $A^T A = \begin{pmatrix}2 & 4\\4 & 8\end{pmatrix}$ is singular. The augmented system row-reduces to $\left(\begin{array}{cc|c}2 & 4 & 7\\0 & 0 & 0\end{array}\right)$, giving infinitely many solutions $\hat{\mathbf{x}} = \begin{pmatrix}7/2 - 2t\\t\end{pmatrix}$. □

Exercise 3.10. $A = \begin{pmatrix}1 & 2\\-1 & 1\\0 & 3\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}1\\3\\0\end{pmatrix}$. (a) Solve via Gram-Schmidt; (b) via normal equations.

Solution. **(a) Gram-Schmidt:** $\mathbf{c}_1 = (1, -1, 0)^T$, $\mathbf{c}_2 = (2, 1, 3)^T$.

$$\mathbf{c}_2^{\perp} = (2, 1, 3)^T - \frac{1}{2}(1, -1, 0)^T = (3/2, 3/2, 3)^T.$$

$$\hat{\mathbf{b}} = \frac{-2}{2}(1, -1, 0)^T + \frac{3 + 9/2 + 0}{9/4 + 9/4 + 9}(3/2, 3/2, 3)^T \approx (-1/3, 5/3, 4/3)^T.$$

Solving $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$ gives $\hat{\mathbf{x}} \approx (-1.222,\ 0.444)^T$.

(b) Normal equations: $A^T A = \begin{pmatrix}2 & -3\\-3 & 14\end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix}-2\\5\end{pmatrix}$. $\det = 19$. $\hat{\mathbf{x}} = \frac{1}{19}\begin{pmatrix}14 & 3\\3 & 2\end{pmatrix}\begin{pmatrix}-2\\5\end{pmatrix} = \frac{1}{19}\begin{pmatrix}-13\\8\end{pmatrix} \approx \begin{pmatrix}-0.684\\0.421\end{pmatrix}$.

Hmm – let me re-check: $A^T A = \begin{pmatrix}1^2 + 1^2 & 1 \cdot 2 + (-1) \cdot 1\\2 \cdot 1 + 1 \cdot (-1) & 2^2 + 1^2 + 3^2\end{pmatrix} = \begin{pmatrix}2 & 1\\1 & 14\end{pmatrix}$. $A^T\mathbf{b} = \begin{pmatrix}1 - 3\\2 + 3\end{pmatrix} = \begin{pmatrix}-2\\5\end{pmatrix}$. $\det = 27$. $\hat{\mathbf{x}} = \frac{1}{27}\begin{pmatrix}14 & -1\\-1 & 2\end{pmatrix}\begin{pmatrix}-2\\5\end{pmatrix} = \frac{1}{27}\begin{pmatrix}-33\\12\end{pmatrix} \approx \begin{pmatrix}-1.222\\0.444\end{pmatrix}$. Both agree. ✓ □

Exercise 3.11. $A = \begin{pmatrix} 2 & -2 \\ 1 & 4 \\ 1 & 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$.

Solution. $A^T A = \begin{pmatrix} 6 & 2 \\ 2 & 24 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 2 \\ 16 \end{pmatrix}$. $\det = 140$. $\hat{\mathbf{x}} = \frac{1}{140}\begin{pmatrix} 24 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} 2 \\ 16 \end{pmatrix} = \frac{1}{140}\begin{pmatrix} 16 \\ 92 \end{pmatrix} \approx \begin{pmatrix} 0.114 \\ 0.657 \end{pmatrix}$.

Error: $\|A\hat{\mathbf{x}} - \mathbf{b}\| \approx 0.507$. □

Exercise 3.12. Explain why a least-squares problem always has a solution.

Solution. $\mathrm{Col}(A)$ is a subspace of $\mathbb{R}^m$. Every point in $\mathbb{R}^m$ has a unique orthogonal projection onto any subspace. So $\hat{\mathbf{b}} = \mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ always exists, $\hat{\mathbf{b}} \in \mathrm{Col}(A)$, and $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$ is consistent. □

Exercise 3.13. What is wrong with $(A^T A)^{-1}A^T = A^{-1}(A^T)^{-1}A^T = A^{-1}$?

Solution. In the least-squares setting, A is typically $m \times n$ with $m > n$ (not square), so A^{-1} **does not exist**. Similarly $(A^T)^{-1}$ does not exist. The factorization $(BC)^{-1} = C^{-1}B^{-1}$ is only valid when both B and C are invertible square matrices. □

Exercise 3.14. Why does switching columns of A not affect the least-squares solution?

Solution. Switching columns multiplies A on the right by a permutation P: $A' = AP$. Since $\mathrm{Col}(A') = \mathrm{Col}(A)$, the projection $\mathrm{Proj}_{\mathrm{Col}(A')}\,\mathbf{b} = \mathrm{Proj}_{\mathrm{Col}(A)}\,\mathbf{b}$ is unchanged, and so is the minimum error. The solution changes as $\hat{\mathbf{x}}' = P^T\hat{\mathbf{x}}$ (reordered to match the new column order), but the residual $A'\hat{\mathbf{x}}' - \mathbf{b} = A\hat{\mathbf{x}} - \mathbf{b}$ is unchanged. □

Chapter 2

Curve Fitting — Solutions

Exercise 4.1. Points $(-1,0),(1,2),(2,2)$: (a) find LS line; (b) what is minimized?

Solution. $A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$. $A^TA = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$, det $= 14$. $\hat{\mathbf{c}} = \frac{1}{14}\begin{pmatrix} 3 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 11/7 \end{pmatrix}$.

LS line: $f(x) = \frac{1}{2}x + \frac{11}{7}$.

(b) The LS line minimizes $\sum(y_i - f(x_i))^2$, the sum of squared *vertical* distances. □

Exercise 4.2. Points $(1,1),(5,2),(6,2),(8,3)$: (a) LS line; (c) estimate y at $x = 20$.

Solution.

```
pts=[1 1;5 2;6 2;8 3]; A=[pts(:,1),ones(4,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$m \approx 0.2692$, $b_0 \approx 0.6538$. **Line:** $f(x) \approx 0.269x + 0.654$.

(c): $f(20) \approx 0.269(20) + 0.654 \approx 6.04$. □

Exercise 4.3. Points $(-1,3),(0,1),(1,2),(3,9)$: (a) LS parabola; (c) solve $f(x) = 10$.

Solution. $A = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 9 & 3 & 1 \end{pmatrix}$.

```
pts=[-1 3;0 1;1 2;3 9]; A=[pts(:,1).^2,pts(:,1),ones(4,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 1.045$, $b \approx -0.555$, $c \approx 1.291$. **Parabola:** $f(x) \approx 1.045x^2 - 0.555x + 1.291$.

(c): $1.045x^2 - 0.555x + 1.291 = 10 \Rightarrow 1.045x^2 - 0.555x - 8.709 = 0$. $x = \frac{0.555 \pm \sqrt{0.308 + 36.41}}{2.09} \Rightarrow$ $x \approx 3.16$ or $x \approx -2.63$. □

Exercise 4.4. Points $(-2,0),(0,3),(4,4)$: LS $f(x) = ae^x + b$; estimate $f'(1)$.

Solution. $A = \begin{pmatrix} e^{-2} & 1 \\ 1 & 1 \\ e^4 & 1 \end{pmatrix}$.

```
pts=[-2 0;0 3;4 4]; A=[exp(pts(:,1)),ones(3,1)]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 0.04693$, $b \approx 1.4615$. **Function:** $f(x) \approx 0.04693e^x + 1.4615$.

$f'(1) = ae \approx 0.04693 \times 2.718 \approx 0.128$. □

Exercise 4.5. Points $(-2, 6.3), (3, 1.2), (5, 7.1), (8, -2.8), (9, -0.05)$: LS $f(x) = a + b \sin x$, estimate $f(\pi/2)$.

Solution.

```
pts=[-2 6.3;3 1.2;5 7.1;8 -2.8;9 -0.05];
A=[ones(5,1),sin(pts(:,1))]; b=pts(:,2);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 2.026$, $b \approx -4.974$.

$f(\pi/2) = 2.026 - 4.974 \sin(\pi/2) = 2.026 - 4.974 \approx -2.948$. □

Exercise 4.6. Points $(-3, -2, 45), (2, -2, 30), (0, 1, 6), (-2, 3, 55), (6, 5, 230)$: LS $f(x, y) = ax^2 + by^2$; estimate $f(3, 5)$.

Solution.

```
pts=[-3 -2 45;2 -2 30;0 1 6;-2 3 55;6 5 230];
A=[pts(:,1).^2,pts(:,2).^2]; b=pts(:,3);
x_hat=inv(A'*A)*A'*b;
```

$a \approx 3.027$, $b \approx 4.818$.

$f(3, 5) \approx 3.027(9) + 4.818(25) \approx 27.24 + 120.45 \approx 147.7$. □

Exercise 4.7. Points $(0, 0), (0, 1), (1, 1)$: find LS line $y = mx + b$ **and** $x = ny + c$. Show they differ.

Solution. **Line** $y = mx + b$**:** $A_1 = \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$, $\mathbf{b}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. $A_1^T A_1 = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$, $A_1^T \mathbf{b}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

$\hat{m} = 1/2$, $\hat{b} = 1/2$. **Line 1:** $y = \frac{1}{2}x + \frac{1}{2}$.

Line $x = ny + c$**:** Swap roles. $\hat{n} = 1$, $\hat{c} = -1/3$. **Line 2:** $x = y - \frac{1}{3}$, i.e., $y = x + \frac{1}{3}$ (slope 1).

Line 1 has slope 1/2; line 2 has slope 1. **Different**, because they minimize different distances (vertical vs. horizontal). □

Exercise 4.8. Orbital data (θ°, r): $(23, 152), (50, 135), (100, 102), (110, 110), (152, 137)$. Fit $Ar^2 \cos^2\theta + Br^2 \sin^2\theta = 1$.

Solution.

```
data=[23 152;50 135;100 102;110 110;152 137];
theta=data(:,1)*pi/180; r=data(:,2);
A=[(r.*cos(theta)).^2,(r.*sin(theta)).^2];
b=ones(5,1); x_hat=inv(A'*A)*A'*b;
```

$A \approx 3.976 \times 10^{-5}$, $B \approx 7.215 \times 10^{-5}$.

(b): At $\theta = 225^\circ$: both $\cos^2, \sin^2 = 1/2$, so $r^2 = 2/(A + B) \approx r \approx 136.5$ Mmiles.

(c): r is maximized when $A \cos^2\theta + B \sin^2\theta$ is minimized. Since $A < B$, minimum at $\theta = 0$: $r_{\max} = 1/\sqrt{A} \approx 158.6$ Mmiles. □

Exercise 4.10. Points $(1, 1), (2, 1), (3, 2), \ldots, (3, 2)$ (n times). Find $\lim_{n\to\infty} \hat{\mathbf{c}}_n$.

Solution. With $n + 2$ points: $A^T A = \begin{pmatrix} 9n + 5 & 3n + 3 \\ 3n + 3 & n + 2 \end{pmatrix}$, $A^T \mathbf{b} = \begin{pmatrix} 6n + 3 \\ 2n + 2 \end{pmatrix}$.

$\det = 5n + 1$. Computing: $m_n = \dfrac{3n}{5n + 1} \to \dfrac{3}{5}$, $b_n = \dfrac{n + 1}{5n + 1} \to \dfrac{1}{5}$.

$\lim_{n\to\infty} \hat{\mathbf{c}}_n = \begin{pmatrix} 3/5 \\ 1/5 \end{pmatrix}$.

Check: $y(3) = \frac{9}{5} + \frac{1}{5} = 2$ ✓ (line passes through $(3, 2)$ in the limit, as expected from infinite weight). □

Exercise 4.13. Which function templates allow LS?

Solution. LS requires all unknowns to appear *linearly*.

(a) $ax^2 + bx^c$: **No** (c is a nonlinear unknown exponent).

(b) $ae^x + bx$: **Yes** (linear in a, b; matrix column i: (e^{x_i}, x_i)).

(c) $e^{ax} + bx$: **No** (a appears in the exponent).

(d) $a \sin x + b \cos x + c$: **Yes** (linear in a, b, c).

(e) $a \sin(bx) + c$: **No** (b appears inside sine).

□

Chapter 3

Team Ranking — Solutions

Exercise 5.1. Four teams with given results. (a) Find rankings. (b) Predict T1 vs T4.

Solution. Games: T1–T2(1), T1–T3(1), T2–T3(2), T2–T4(1), T3–T4(1). Point totals: $q_1 = -1$, $q_2 = 13$, $q_3 = -8$, $q_4 = -4$ (check: sum $= 0$ ✓).

$$M = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 4 & -2 & -1 \\ -1 & -2 & 4 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \bar{q} = \begin{pmatrix} -1 \\ 13 \\ -8 \\ 0 \end{pmatrix}.$$

Rankings: $r_1 = 0.125$, $r_2 = 2.375$, $r_3 = -1.125$, $r_4 = -1.375$. Order: T2>T1>T3>T4.

(b): Expected margin $= r_1 - r_4 = 1.5$. T1 expected to win by 1.5. □

Exercise 5.2. Four teams. (a) Find rankings. (b) Predict T2 vs T4.

Solution. Same M as 5.1 (same game counts). $\bar{q} = (38, -21, -23, 0)^T$. **Rankings:** $r_1 = 13.5$, $r_2 = -5.333$, $r_3 = -5.667$, $r_4 = -2.5$. Order: T1>T4>T2≈T3.

(b): $r_2 - r_4 = -2.833$. T4 expected to win by about 2.8. □

Exercise 5.3. Read off structural info from a given Massey equation.

Solution. $M = \begin{pmatrix} 3 & -1 & -2 & 0 \\ -1 & 4 & -2 & -1 \\ -2 & -2 & 4 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$.

(a) 4 teams. **(b)** T4 original diagonal entry: $|m_{42}| = 1$, so 1 game (vs T2). **(c)** T1: $m_{11} = 3$ games; $|m_{12}| = 1$ vs T2, $|m_{13}| = 2$ vs T3 (twice). **(d)** Original $m_{44} = 1$. Total $= (3 + 4 + 4 + 1)/2 = 6$ games. □

Exercise 5.4. Show the sum of diagonal entries of M (before modification) is always even.

Solution. $\sum_i m_{ii} = \sum_i (\text{games team } i \text{ played}) = 2G$ (each game is counted in exactly 2 teams' totals). Since $2G$ is even, done. □ □

Exercise 5.5. Four teams with home advantage. (a) Without home, (b) reduce home wins by 25%.

Solution. **(a):** $q_1 = 15, q_2 = -22, q_3 = 14, q_4 = -7$. Rankings: $r_1 = 6.5, r_2 = -4.0, r_3 = 2.0, r_4 = -4.5$.

(b): Reduce home wins: T1 at home beats T2 by $5 \to 3.75$; T3 at home beats T4 by $21 \to 15.75$. $q_1 = 3.75 + 10 = 13.75$, $q_2 = -3.75 - 10 + 7 - 14 = -20.75$, $q_3 = -10 - 7 + 15.75 = -1.25$,

$q_4 = 14 - 15.75 = -1.75$. Rankings (b): $r_1 \approx 4.125, r_2 \approx -4.375, r_3 \approx -1.125, r_4 \approx 1.375$. T4 moved above T3. □

Exercise 5.6. If T1 beats T2 by k, T2 beats T3 by k, T3 beats T1 by k: show equal rankings.

Solution. $q_1 = k - k = 0$, $q_2 = -k + k = 0$, $q_3 = -k + k = 0$. With $\bar{q} = \mathbf{0}$ and the constraint row, the unique solution is $\hat{\mathbf{r}} = \mathbf{0}$. All teams have equal ranking 0. □ □

Exercise 5.9 / 5.10. Better to beat team twice by 1 or once by 2?

Solution. **(a) Once by 2:** T1-T2(1), T2-T3(1), T1-T3(1) with margins 3, 2, 2. $M = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (5, -1, 0)^T$. Rankings: $r_1 \approx 1.667$.

(b) Twice by 1: T1-T3 now twice (total margin 2). $M = \begin{pmatrix} 3 & -1 & -2 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (5, -1, 0)^T$. Rankings: $r_1 \approx 1.067$.

One win by 2 gives T1 a higher ranking than two wins by 1 each. □

Exercise 5.11. (a) T1 beats T2 by n, T2 beats T3 by 1, T3 beats T1 by 1. Find $\lim_{n\to\infty} \hat{\mathbf{r}}_n$. (b) T1 beats T2 by 1 n times instead.

Solution. **(a):** $q_1 = n - 1$, $q_2 = -(n-1)$, $q_3 = 0$. By symmetry $r_3 = 0$, $r_1 = -r_2$. Row 1: $3r_1 = n - 1$, so $r_1 = (n-1)/3 \to \infty$.

(b): Now $m_{11} = n + 1$, $m_{12} = m_{21} = -n$. Same point totals but $(2n+1)r_1 = n - 1$, so $r_1 = (n-1)/(2n+1) \to 1/2$.

Winning once by n is unlimited; winning n times by 1 converges to a bounded limit. □

Exercise 5.12. T1 beats T2 by α, T2 beats T3 by β, T3 beats T1 by γ. When are all rankings equal?

Solution. $q_1 = \alpha - \gamma$, $q_2 = -\alpha + \beta$, $q_3 = -\beta + \gamma$. All zero $\Leftrightarrow \alpha = \beta = \gamma$.

Two teams can tie (e.g., $\alpha = 2, \beta = 4, \gamma = 0$ gives $r_1 = r_3$). Spread is unbounded as $\alpha \to \infty$. □

Exercise 5.13. Two groups $\{T1, T2, T3\}$ and $\{T4, T5, T6\}$ with no inter-group games. (a) Massey fails; (b) rank groups; (c) after T3 beats T4 by 1.

Solution. **(a):** Two disconnected components $\Rightarrow$ block-singular Massey matrix.

(b): Each group has $M_{\text{sub}} = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 1 & 1 & 1 \end{pmatrix}$, $\bar{q} = (2, 1, 0)^T$. Rankings: $r_1 \approx 2.333, r_2 \approx 0.333, r_3 \approx -2.667$ (same for T4,T5,T6).

```
M6=[1 -1 0 0 0 0;-1 2 -1 0 0 0;0 -1 2 -1 0 0;
    0 0 -1 2 -1 0;0 0 0 -1 2 -1;1 1 1 1 1 1];
q6=[2;-2+3;-3+1;-1+2;-2+3;0]; r6=M6\q6;
```

(c) Output: $r_1 \approx 5.33, r_2 \approx 3.33, r_3 \approx 0.33, r_4 \approx -0.67, r_5 \approx -2.67, r_6 \approx -5.67$. □

Exercise 5.16. Nontrivial examples where no LS is needed; is sum constraint still required?

Solution. **Example:** T1 beats T2 by 3, T2 beats T3 by 3, T1 beats T3 by 6. The three equations $r_1 - r_2 = 3$, $r_2 - r_3 = 3$, $r_1 - r_3 = 6$ are consistent, so the original $A\mathbf{r} = \mathbf{p}$ has an exact solution.

Sum constraint still needed: Even with consistent results, the Massey matrix $A^T A$ is singular (its rows sum to 0). Without $\sum r_i = 0$, there are infinitely many solutions differing by a constant shift. The constraint gives the unique centered solution. $\square$

Chapter 4

Graph Theory — Solutions

Exercise 12.1. Graph: 1–2, 1–3, 2–3, 3–4. Count walks and explain.

Solution.

```
A=make_adj([1 2;1 3;2 3;3 4],4); fprintf('%d\n',A^3(1,3));
```

(a) $(A^3)_{13} = 4$ walks of length 3 from v1 to v3.

(b) $(A^{20})_{24} = 790748$.

(c) $(A^3)_{44} = 0$. Vertex 4 has degree 1 (only neighbor: v3). A closed walk v4→v3→?→v4 of length 3 would need to return from a neighbor of v3 to v4 in one step; but neighbors of v3 in $\{v1, v2, v4\}$ can only reach v4 via v3, which already used one step. No such walk exists. □

Exercise 12.2. Cycle C_5: 1–2–3–4–5–1. Count walks and identify patterns.

Solution. $(A^3)_{12} = 3$. $(A^{10})_{24} = 220$.

(c) Even walks $v3 \to v5$: $k = 2 : 1,\ k = 4 : 4,\ k = 6 : 15,\ k = 8 : 57,\ k = 10 : 220$. Growing because $v3$ and $v5$ are 2 apart and more paths accumulate with length.

(d) Odd walks $v2 \to v4$: $k = 1 : 0,\ k = 3 : 1,\ k = 5 : 5,\ k = 7 : 22,\ k = 9 : 93$. Growing similarly (v2 and v4 are 2 apart going one way, 3 the other). □

Exercise 12.3. Why does $\frac{1}{6}\operatorname{tr}(A^3)$ count triangles but $c \cdot \operatorname{tr}(A^4)$ doesn't count squares?

Solution. Every closed walk of length 3 in a simple graph is a triangle (no shortcuts possible). So $\operatorname{tr}(A^3)$ counts triangle traversals exactly.

Closed walks of length 4 include squares *and* backtracks ($v \to u \to v \to u \to v$) and triangle-with-backtrack paths. $\operatorname{tr}(A^4)$ therefore overcounts. The same issue applies to all cycle lengths ≥ 4. □

Exercise 12.4. Apply Fiedler to graph 1–2, 1–3, 2–3, 3–4.

Solution. $L = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$.

```
A=make_adj([1 2;1 3;2 3;3 4],4); L=make_lap(A);
[fv,lam2]=fiedler_vec(L);
```

Eigenvalues: $0, 1, 3, 4$. Fiedler value $\lambda_2 = 1$. Fiedler vector $\approx (-0.408, -0.408, 0, 0.816)^T$.

Partition: $V_- = \{1, 2\}$, $V_0 = \{3\}$ (bridge), $V_+ = \{4\}$. Cut = 1 edge (3–4). □

Exercise 12.5. Apply Fiedler to graph with two triangles sharing vertex 3.

Solution. Graph: 1–2, 1–3, 2–3, 3–4, 3–5, 4–5.

Fiedler vector $\approx (-0.5, -0.5, 0, 0.5, 0.5)^T$. Partition: $V_- = \{1, 2\}$, $V_+ = \{4, 5\}$. Vertex 3 is the bridge (can go in either group). □

Exercise 12.6. Fiedler vector $(-0.195, -0.632, -0.195, 0.512, 0.512)^T$: partition.

Solution. $V_- = \{1, 2, 3\}$ (negative entries), $V_+ = \{4, 5\}$ (positive entries). Partition: $(\{1, 2, 3\}, \{4, 5\})$. □

Exercise 12.7. (Requires the specific graph from the notes – Fiedler computation.)

Solution.

```
% Replace edge list below with the specific graph from Exercise 12.7
A=make_adj([...],n); L=make_lap(A); [fv,lam2]=fiedler_vec(L);
neg=find(fv<0); pos=find(fv>0);
```

Apply algorithm: build L, find Fiedler vector, partition by sign. □

Exercise 12.8. Match graphs G1, G2, G3 to Fiedler vectors $\bar{v}, \bar{w}, \bar{x}$.

Solution. G2 (linear chain): Fiedler vector is monotone $\to$ matches $\bar{v} = (-0.56, -0.41, -0.15, 0.15, 0.41, 0.56)$.

G1 (symmetric 2-wing): Two equal clusters $\to$ matches $\bar{w} = (0.46, 0.46, 0.26, -0.26, -0.46, -0.46)$.

G3 (asymmetric): matches $\bar{x} = (0.32, 0.51, 0.32, -0.12, -0.51, -0.51)$. □

Exercise 12.9. (Specific graph from notes – Fiedler method.)

Solution. Use the MATLAB helper functions on the specific adjacency structure of exercise 12.9 (as given in the text). The Fiedler vector partitions the graph by sign. □

Exercise 12.13. K_n: what are the eigenvalues of the Laplacian?

Solution. $L_{K_n} = nI - J$ where $J = \mathbf{1}\mathbf{1}^T$. Eigenvectors of J: $\mathbf{1}$ (eigenvalue n) and any vector orthogonal to $\mathbf{1}$ (eigenvalue 0). So eigenvalues of $nI - J$: $n - n = 0$ (once) and $n - 0 = n$ ($n - 1$ times).

Verification:

```
for n=[3 4 5 6], A=ones(n,n)-eye(n); L=diag(sum(A))-A;
fprintf('K_%d: ',n); fprintf('%.1f ',sort(eig(L))); fprintf('\n'); end
```

□

Exercise 12.18. LAN with 10 computers: Laplacian, Fiedler vector, partition.

Solution.

```
edges=[1 8;1 9;1 10;2 5;2 6;3 4;3 9;4 9;5 7;6 7;6 8;8 10];
A=make_adj(edges,10); L=make_lap(A); [fv,lam2]=fiedler_vec(L);
```

Fiedler value ≈ 0.1755.

Partition: $\{C1, C3, C4, C9, C10\}$ vs $\{C2, C5, C6, C7, C8\}$. This separates the C1/C9-hub cluster from the C2/C6-hub cluster. Cut = number of edges between groups. □

Exercise SP3 Exam: Graph Isomorphism. (a) How many 3×3 permutation matrices? (b) Check $PA_2P^T = A_1$ for all. (c) Use triangle count.

Solution. **(a)** $3! = 6$.

(b) $A_1 = K_3$ (triangle), $A_2 = P_3$ (path). None of the 6 permutations gives $PA_2P^T = A_1$.

```
A1=[0 1 1;1 0 1;1 1 0]; A2=[0 1 0;1 0 1;0 1 0];
for k=1:6, P=eye(3)(perms([1 2 3])(k,:),:);
  if all(all(P*A2*P'==A1)), disp('equal'); end; end
```

Not isomorphic.

(c) Triangles(K_3)= $\mathrm{tr}(A_1^3)/6 = 1$; Triangles(P_3)= $\mathrm{tr}(A_2^3)/6 = 0$. Different $\Rightarrow$ not isomorphic. □ □

True/False:

"Same adjacency matrix $\Rightarrow$ isomorphic": **True**. Identical adjacency matrices (with the same vertex labeling) means the graphs have exactly the same edge structure, hence $P = I$ witnesses isomorphism. □

"Same degree matrix $\Rightarrow$ isomorphic": **False**. Counterexample: C_6 (6-cycle, every vertex degree 2) and $2K_3$ (two disjoint triangles, every vertex degree 2) both have degree sequence $(2, 2, 2, 2, 2, 2)$ but are not isomorphic (C_6 is connected, $2K_3$ is not).

Chapter 5

Markov Chains — Solutions

Exercise 6.1. Population diagram (3 states): (a) transition matrix; (b) asymmetry; (c) 1 and 2 iterations; (d) steady state; (e) smallest k for 4-decimal accuracy.

Solution. $T = \begin{pmatrix} 0.80 & 0.05 & 0.25 \\ 0.12 & 0.90 & 0.05 \\ 0.08 & 0.05 & 0.70 \end{pmatrix}$, $\mathbf{x}_0 = (0.6, 0.2, 0.2)^T$.

(b) T not symmetric $\Rightarrow$ movement rates differ by direction.

(c) $\mathbf{x}_1 = T\mathbf{x}_0 = (0.540, 0.262, 0.198)^T$; $\mathbf{x}_2 = T^2\mathbf{x}_0 \approx (0.497, 0.303, 0.200)^T$.

(d) Solve $(T - I)\mathbf{x}^* = \mathbf{0}$ with $\sum x_i = 1$: $\mathbf{x}^* \approx (0.416, 0.416, 0.168)^T$.

(e)

```
T=[0.80 0.05 0.25;0.12 0.90 0.05;0.08 0.05 0.70];
x=T^1*[0.6;0.2;0.2]; xs=steady_state(T); k=1;
while max(abs(x-xs))>5e-5, k=k+1; x=T*x; end; fprintf('k=%d\n',k);
```

□

Exercise 6.2. 6×6 transition matrix: (a) diagram; (b) regular?

Solution. $T = \begin{pmatrix} 0.9 & 0 & 0.05 & 0 & 0.15 & 0.05 \\ 0 & 0.8 & 0.05 & 0.2 & 0.05 & 0.05 \\ 0.01 & 0.1 & 0.5 & 0 & 0 & 0.05 \\ 0.02 & 0.1 & 0.1 & 0.75 & 0 & 0.05 \\ 0.07 & 0 & 0.1 & 0 & 0.7 & 0.05 \\ 0 & 0 & 0.2 & 0.05 & 0.1 & 0.75 \end{pmatrix}$.

(b) Regular, since T^2 has all positive entries. All states communicate (every state can reach every other state in at most 2 steps). □

Exercise 6.3. Construct a 3-state regular symmetric transition matrix.

Solution. $T = \begin{pmatrix} 0.6 & 0.2 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{pmatrix}$.

Symmetric (equal migration rates), regular (all entries positive). Steady state: $\mathbf{x}^* = (1/3, 1/3, 1/3)^T$ by symmetry. □

Exercise 6.4. Large diagram: transition matrix, entries of T^k, zero entries for all k.

Solution. T is block upper-triangular (states 4,5,6 cannot reach states 1,2,3).

(a) Entry $(3, 2)$ of T^1: read directly from diagram. **(b)** Entry $(3, 2)$ of T^2: $\sum_\ell T_{3\ell} T_{\ell 2}$. **(c)** Entry $(1, 3)$ of T^2: 0 if 3 cannot reach 1 in 2 steps. **(d)** Smallest k for nonzero $(5, 3)$: path length

$3 \to 4 \to 5$ or $3 \to 5$. **(e)** (i, j) zero for all k: if state j is in the "absorbing" block unreachable from i *or* vice versa. **(f)** Long term: population drains into states 4,5,6. □

Exercise 6.5. Transition matrix with $T^7 = I$ but $T^k \neq I$ for $k < 7$.

Solution. Use a 7-state chain cycling $1 \to 2 \to 3 \to \cdots \to 7 \to 1$ deterministically: $T_{i+1,i} = 1$ (and $T_{1,7} = 1$), 0 elsewhere. Then $T^7 = I$. This is not regular (period 7). □

Exercise 6.6. 5×5 non-regular T with all states converging to a unique steady state.

Solution. States $\{1, 2, 3\}$ form a regular sub-chain; states 4,5 always transition into $\{1, 2, 3\}$. Example:

$$T = \begin{pmatrix} 0.7 & 0.1 & 0.2 & 0.5 & 0.5 \\ 0.1 & 0.8 & 0.1 & 0.3 & 0.3 \\ 0.2 & 0.1 & 0.7 & 0.2 & 0.2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Non-regular (states 4,5 are absorbing-like and $T_{i4} = T_{i5} = 0$ for $i > 3$). But all states eventually drain to $\{1, 2, 3\}$, which has a unique steady state. □

Exercise 6.7. 3-state diagram: T, iterations, steady state.

Solution. $T = \begin{pmatrix} 0.70 & 0.05 & 0.02 \\ 0.28 & 0.85 & 0.03 \\ 0.02 & 0.10 & 0.95 \end{pmatrix}$, $\mathbf{x}_0 = (0.1, 0, 0.9)^T$.

(a) Transition matrix: as above.

(b) $\mathbf{x}_1 = T\mathbf{x}_0 = (0.088, 0.113, 0.799)^T$; compute further iterations programmatically.

(c) Steady state: solve $(T - I)\mathbf{x}^* = \mathbf{0}$, $\sum x_i = 1$.

```
T=[0.70 0.05 0.02;0.28 0.85 0.03;0.02 0.10 0.95];
x0=[0.1;0;0.9]; [V,D]=eig(T); idx=find(abs(diag(D)-1)<1e-9);
xs=real(V(:,idx)); xs=abs(xs)/sum(abs(xs));
```

□

Exercise 6.8. Find eigenvalues of $T = \begin{pmatrix} a & 0 & 1-c \\ 1-a & b & 0 \\ 0 & 1-b & c \end{pmatrix}$ (columns sum to 1).

Solution. $\lambda = 1$ is always an eigenvalue (columns sum to 1). The characteristic polynomial gives:

$$\mathrm{char}(T) = (\lambda - 1)\big(\lambda^2 - (a + b + c - 1)\lambda + (ab + bc + ca - a - b - c + 1)\big).$$

The other two eigenvalues are roots of the quadratic factor. Their values depend on a, b, c. □

Exercise 6.9. Find a 2×2 regular T and $\mathbf{x}_0$ where $\mathbf{x}_{1000}$ still differs from $\mathbf{x}^*$ at the first decimal place.

Solution. Slow convergence requires $|\lambda_2| = |a+b-1|$ close to 1. Take $a = 0.9999$ and $b = 0.9999$:

$$T = \begin{pmatrix} 0.9999 & 0.0001 \\ 0.0001 & 0.9999 \end{pmatrix}.$$

$\lambda_2 = 0.9998$. $\lambda_2^{1000} = 0.9998^{1000} \approx 0.819$. Deviation is $\approx 0.5 \times 0.819 \approx 0.41$, so first decimal place still differs. $\mathbf{x}_0 = (1, 0)^T$, $\mathbf{x}^* = (0.5, 0.5)^T$. □

Exercise 6.10. Transition matrix for a disconnected diagram.

Solution. Two disconnected 3-state chains:

$$T = \begin{pmatrix} 0.7 & 0.1 & 0.2 & 0 & 0 & 0 \\ 0.1 & 0.8 & 0.1 & 0 & 0 & 0 \\ 0.2 & 0.1 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.6 & 0.3 & 0.1 \\ 0 & 0 & 0 & 0.2 & 0.5 & 0.3 \\ 0 & 0 & 0 & 0.2 & 0.2 & 0.6 \end{pmatrix}.$$

□

Exercise 6.11. Disconnected T: one component with steady state, one without.

Solution. Component 1: regular (has steady state). Component 2: $T_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (non-regular, oscillates). Block-diagonal:

$$T = \begin{pmatrix} T_1 & 0 \\ 0 & T_2 \end{pmatrix}.$$

□

Exercise 6.12. Define: probability vector, transition matrix, regular transition matrix.

Solution. **Probability vector:** A non-negative vector whose entries sum to 1.

Transition (stochastic) matrix: A matrix whose columns are all probability vectors. Entry T_{ij} = fraction of population moving from state j to state i per step.

Regular transition matrix: A transition matrix T such that T^k has all strictly positive entries for some integer $k \geq 1$. □

Exercise 6.13. For any n: can some entry of T^k be 0 for $k < n$ but nonzero at $k = n$?

Solution. **Yes.** Use the n-cycle transition matrix: $T_{i+1,i} = 1$ cyclically. Then $(T^k)_{1,1} = 0$ for $1 \leq k < n$ (no closed walk of length $< n$ starting and ending at state 1), but $(T^n)_{1,1} = 1$. □

Exercise 6.14. Two-state: (a) transition matrix; (b) steady state; (c) restrictions.

Solution. $T = \begin{pmatrix} \alpha & \beta \\ 1-\alpha & 1-\beta \end{pmatrix}$ where α = stay at 1, $1-\beta$ = stay at 2.

(b) Steady state: $(T - I)\mathbf{x} = \mathbf{0}$, $x_1 + x_2 = 1$:

$$x_1^* = \frac{\beta}{(1-\alpha)+\beta}, \quad x_2^* = \frac{1-\alpha}{(1-\alpha)+\beta}.$$

(c) If $\alpha = 1, \beta = 1$ (no movement): $T = I$, any probability vector is a steady state (∞ many). If $\alpha = 0, \beta = 0$: $T = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$, state 2 is absorbing – unique steady state $\mathbf{x}^* = (0,1)^T$ but not "limiting" from all starts. Without the restrictions, steady states can be non-unique or non-convergent. □

Exercise 6.15. If T^k has all nonzero entries, show T^j does too for $j > k$.

Solution. Since T is a transition matrix, all entries $T_{ij} \geq 0$ and each column sums to 1 (so at least one entry per column is > 0). For $j = k+1$: $(T^{k+1})_{ij} = \sum_\ell T_{i\ell}(T^k)_{\ell j}$. Since $(T^k)_{\ell j} > 0$ for all ℓ, j and $T_{i\ell} \geq 0$ with $\sum_\ell T_{i\ell} = 1 > 0$, at least one term is positive. So $(T^{k+1})_{ij} > 0$. By induction, T^j has all positive entries for all $j \geq k$. □ □

Chapter 6

Google PageRank — Solutions

Exercise 7.1. Build the PageRank transition matrix for a small internet and find the ranking vector.

Solution. Example: P1→P2, P1→P3, P2→P3, P3→P1. Link matrix:

$$S = \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{pmatrix}.$$

Google matrix with $d = 0.85$, $n = 3$:

$$T = \frac{0.15}{3} J + 0.85 S = \begin{pmatrix} 0.05 & 0.05 & 0.9 \\ 0.475 & 0.05 & 0.05 \\ 0.475 & 0.9 & 0.05 \end{pmatrix}.$$

```
links=[1 2;1 3;2 3;3 1]; T=make_gpmatrix(links,3); pr=pagerank_vec(T);
```

Output: $\mathbf{x}^* \approx (0.278, 0.155, 0.567)^T$. Ranking: P3 > P1 > P2. □

Exercise 7.2. Construct the PageRank matrix from a web diagram with 4 pages; rank them.

Solution.

```
links=[1 2;2 1;2 3;3 4;4 1;4 2]; T=make_gpmatrix(links,4); pr=pagerank_vec(T);
```

Rankings depend on the diagram. The page receiving the most links from high-ranked pages wins. □

Exercise 7.3. How does adding a dangling node affect the matrix?

Solution. A dangling node j (no outbound links) contributes column $j = (1/n, \ldots, 1/n)^T$ to S – a uniform jump. The PageRank matrix is unchanged in structure; the dangling node merely distributes its "page traffic" equally. Adding dangling nodes slightly dilutes the link-structure information but the matrix remains regular. □

Exercise 7.4. $P1 \leftrightarrow P2$: both have equal PageRank? Prove by symmetry.

Solution. $T = \frac{0.15}{2} J + 0.85 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0.075 & 0.925 \\ 0.925 & 0.075 \end{pmatrix}$.

Steady state: $T\mathbf{x}^* = \mathbf{x}^*$. Symmetry gives $x_1^* = x_2^* = 0.5$. □

Exercise 7.5. Can adding outbound links change your PageRank?

Solution. **No.** Outbound links determine *where your traffic goes*, not how much you receive. Adding an outbound link to page j changes column j of S (and hence T), but this changes the steady state *of other pages*, not necessarily yours.

In practice, adding outbound links slightly reduces your traffic concentration (your column sends traffic elsewhere), which can reduce your PageRank marginally. □

Exercise 7.6. Chain internet $P1 \to P2 \to P3$ (P3 dangling): build T, rank.

Solution. $n = 3$. P3 is dangling, so $S_{j3} = 1/3$ for all j:

$$S = \begin{pmatrix} 0 & 0 & 1/3 \\ 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \end{pmatrix}, \quad T = \frac{0.15}{3} J + 0.85S.$$

```
links=[1 2;2 3]; T=make_gpmatrix(links,3); pr=pagerank_vec(T);
```

Ranking: P3>P2>P1 (P3 receives a link from P2 which receives from P1; traffic accumulates further down the chain, plus P3 redistributes uniformly). □

Exercise 7.7. Effect of damping factor d on rankings.

Solution. As $d \to 1$: $T \to S$ (pure link structure). Isolated pages with no inbound links die; all traffic follows links exactly. Unique steady state may not exist if S is not regular.

As $d \to 0$: $T \to \frac{1}{n}J$. $T\mathbf{x} = \frac{1}{n}J\mathbf{x} = \frac{1}{n}\mathbf{1} \to \mathbf{u} = (1/n, \ldots, 1/n)^T$. Uniform – link structure has no effect.

$d = 0.85$ is empirically chosen to weight link structure heavily while guaranteeing convergence. □

Exercise 7.8. Fully connected n-page internet: show $\mathbf{u} = (1/n, \ldots, 1/n)^T$ is the steady state.

Solution. Each page links to all $n-1$ others, so $S_{ij} = 1/(n-1)$ for $i \neq j$ and $S_{ii} = 0$. The all-ones matrix J gives $J\mathbf{u} = \mathbf{1}$. Then:

$$T\mathbf{u} = \frac{0.15}{n} J\mathbf{u} + 0.85S\mathbf{u} = \frac{0.15}{n} \cdot n\mathbf{u} + 0.85 \cdot \frac{n}{n-1}\mathbf{1} \cdot \frac{1}{n}.$$

Wait: $S\mathbf{u} = \frac{1}{n-1}(J - I)\mathbf{u} = \frac{1}{n-1}(\mathbf{1} - \mathbf{u}) = \frac{1}{n-1} \cdot \frac{n-1}{n}\mathbf{1} = \frac{1}{n}\mathbf{1} = \mathbf{u}$. So $T\mathbf{u} = 0.15\mathbf{u} + 0.85\mathbf{u} = \mathbf{u}$. ✓
□ □

Exercise 7.9. Which of the following is more effective at raising your PageRank: (a) getting a link from a high-ranked page, or (b) getting many links from low-ranked pages?

Solution. **(a) One high-ranked link is generally better.** A high-ranked page passes a large share of its traffic. A low-ranked page passes very little. Even many low-ranked inbound links may not equal one from a page that itself has high PageRank.

However, there's a trade-off: if a high-ranked page links to many pages, each gets only a small share ($1/k$ of the high rank). So the answer also depends on how many other outbound links the high-ranked page has. □

Exercise 7.10. Disconnected internet without random jump: why fails? Fix?

Solution. Without the random-jump term: $T = S$. If the internet is disconnected, S is block upper-triangular (or has structural zeros). S may not be regular $\Rightarrow$ no unique steady state $\Rightarrow$ PageRank is not uniquely defined.

Fix: The random-jump term $\frac{1-d}{n}J$ adds $(1-d)/n > 0$ to every entry, making T strictly positive. Hence T^1 has all positive entries $\Rightarrow$ T is regular $\Rightarrow$ unique steady state guaranteed. □

Exercise 7.11. Build and rank a 6-page internet with the given link structure.

Solution.

```
links=[1 2;1 4;2 3;3 1;4 5;5 6;6 4;2 5];
T=make_gpmatrix(links,6,0.85); pr=pagerank_vec(T);
fprintf('PageRank:␣'); fprintf('%.4f␣',pr); fprintf('\n');
[~,order]=sort(pr,'descend');
fprintf('Ranking:␣'); fprintf('P%d␣',order); fprintf('\n');
```

Output: Ranking varies by link structure (output shown in compiled code). □